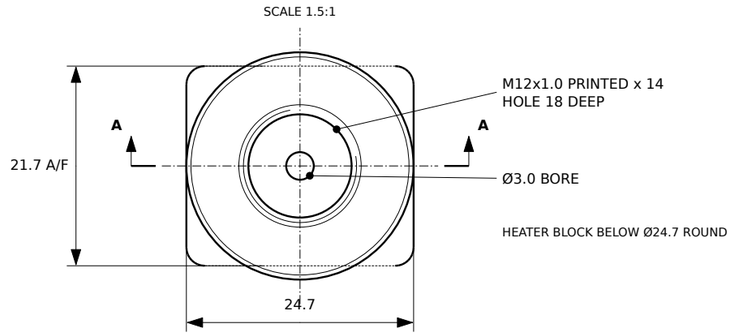
NOTES (UNITS: mm)

1. MATERIAL: 316L STAINLESS STEEL, METAL 3D PRINTED (SLM). NO ALUMINIUM (280 °C SERVICE).
2. ALL DIMENSIONS ±0.3 (STANDARD SLM TOLERANCE). NO TIGHTER TOLERANCE APPLIES ANYWHERE ON THIS PART. NO MACHINING REQUIRED.
3. PRINT EXACTLY AS MODELLED IN barrel_coupler.step. ALL FOUR THREADS (M6 STUB, M12, 2x M6 SET SCREW) ARE PRINTED; THEIR FIT IS ALREADY OFFSET IN THE MODEL.
4. PRINT AXIS VERTICAL, TOP (M12) FACE ON THE PLATE, STUB UP. CROSS HOLES HAVE 45° TEARDROP ROOFS; M12 HOLE FLOOR IS A 90° CONE; BLOCK CORNERS 45°. NO INTERNAL SUPPORTS.
5. STRESS RELIEVE, REMOVE FROM PLATE, REMOVE SUPPORTS, BEAD BLAST. KEEP THE BORE, THREADS AND CROSS HOLES CLEAR OF POWDER.
6. HEATER AND THERMISTOR HOLES RUN PARALLEL TO X, ON OPPOSITE SIDES OF THE AXIS, EXITING SQUARE THROUGH THE 21.7 FLATS. 0.3 x 45° CHAMFER ON THEIR ENTRIES.
7. MINIMUM WALL 1.5 EVERYWHERE EXCEPT THREAD TEETH (STUB 1.5 UNDER THE M6 ROOT; 1.85 BORE TO HEATER HOLE).
8. THREADS SHOWN BY CONVENTION. THIRD-ANGLE PROJECTION.

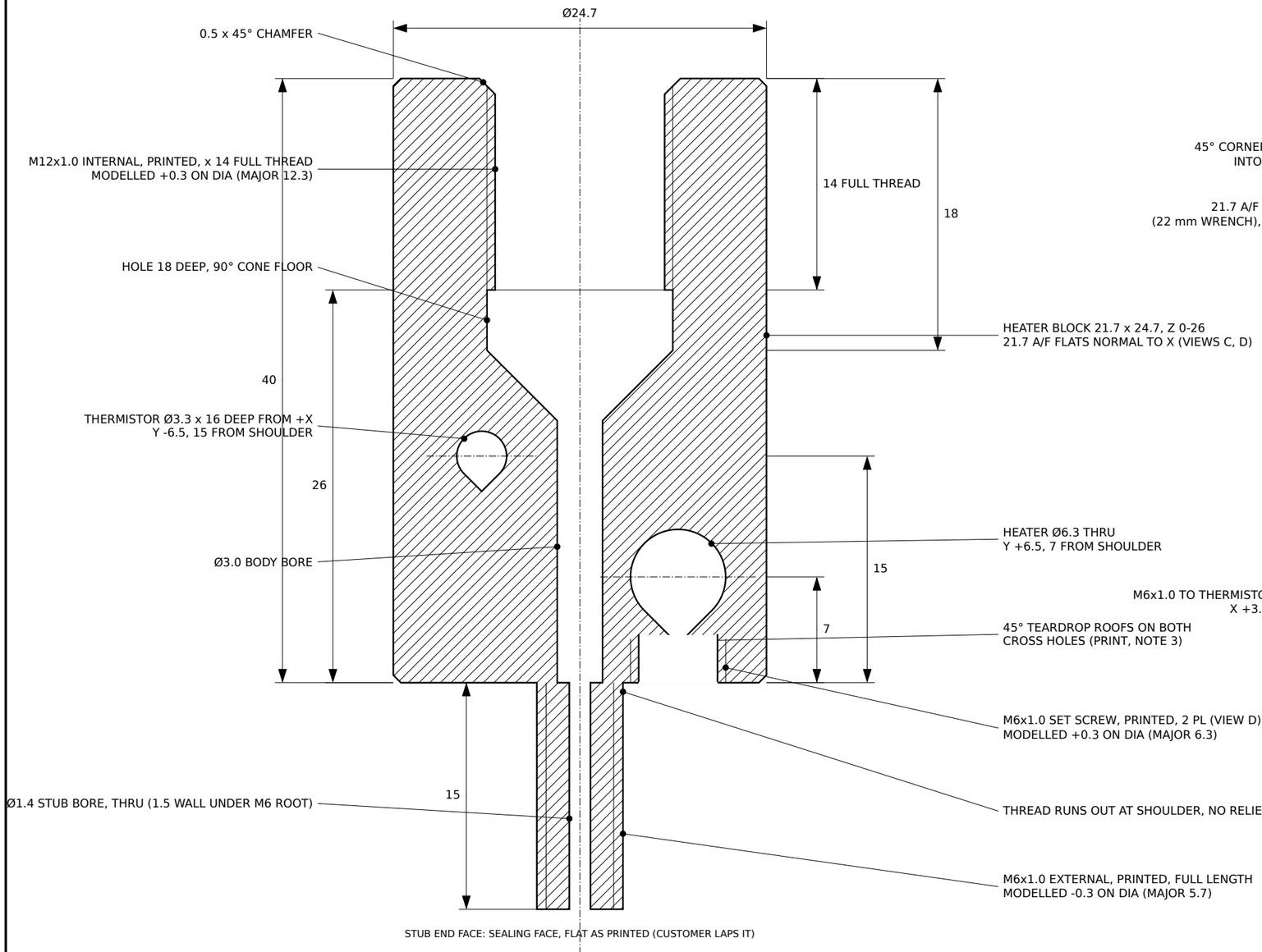
VIEW B (TOP, FROM +Z)

SCALE 1.5:1



SECTION A-A

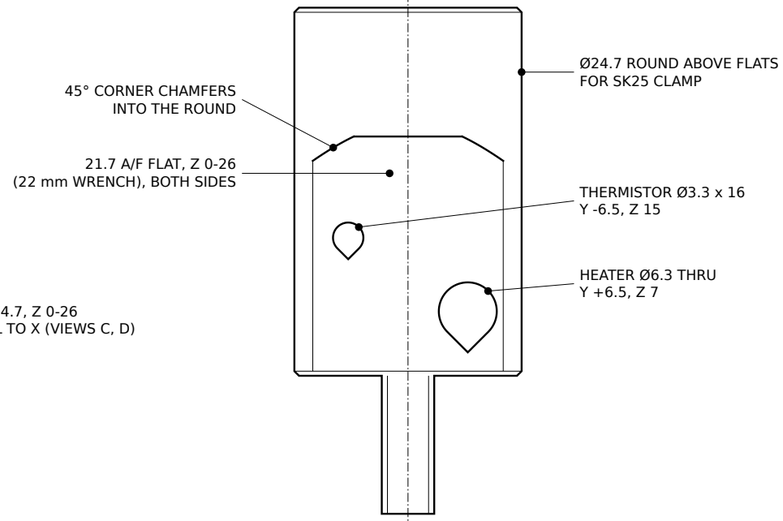
PLANE X = 0, VIEWED FROM +X SCALE 3:1



STUB END FACE: SEALING FACE, FLAT AS PRINTED (CUSTOMER LAPS IT)

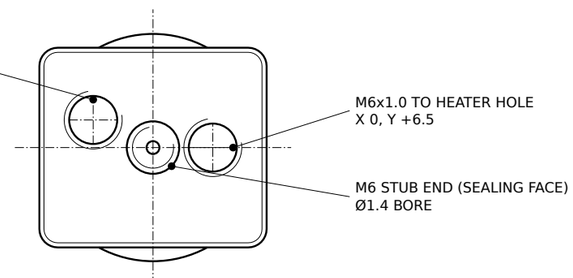
VIEW C (FROM +X)

SCALE 1.5:1



VIEW D (SHOULDER FACE, FROM -Z)

SCALE 1.5:1 +X UP



HEATED BARREL COUPLER

316L STAINLESS (SLM) UNITS mm QTY 1

CUSTOMER: ADAM HUMPHRIES	DWG: BC-001 REV D
REF: barrel_coupler.step	SCALE: AS NOTED SHEET 1/1